HW2 - Pham Minh Trí - 2353235

A1. Impurity intuition before calculation.

- Ranking: (12,0) - pure node, no uncertainty < (9,3) - 75%|25%, in between < (6,6) - a coin flip, most uncertain.
- Check with entropy $H = -\sum p_i \log_2 p_i$ (with $0 \log_2 0 = 0$)

$H(12,0) = -(1 \log_2 1 + 0) = 0$

$H(9,3) = -(0.75 \log_2 0.75 + 0.25 \log_2 0.25) \approx 0.811$

$H(6,6) = -2(0.5 \log_2 0.5) = 1$

$0 < 0.811 < 1 \Rightarrow$ the rank is confirmed : $(12,0) < (9,3) < (6,6)$

A2. Compare two ~~candidates~~ candidate splits with Gini.

$G = 1 - \sum_k p_k^2$. Parent (11P, 7N), $N = 18$: $G = 1 - \frac{11^2 + 7^2}{18^2} \approx 0.4753$

Split S1 : children (6P, 1N) & (5P, 6N).

$G(6,1) = 1 - \frac{6^2 + 1^2}{7^2} \approx 0.2449$ | $G(5,6) = 1 - \frac{5^2 + 6^2}{11^2} \approx 0.4959$

$G_{S1} = \frac{7}{18} \times 0.2449 + \frac{11}{18} \times 0.4959 \approx 0.3983$

Split S2 : children (8P, 4N) & (3P, 3N)

$G(8,4) = 1 - \frac{8^2 + 4^2}{12^2} = \frac{4}{9} \approx 0.4444$ | $G(3,3) = 0.5$

$\Rightarrow G_{S2} = \frac{12}{18} \times \frac{4}{9} + \frac{6}{18} \times 0.5 \approx 0.4630 > 0.3983$ (G_{S1})

$\Rightarrow$ CART prefers S1. Impurity decrease: $0.4753 - 0.3983 = 0.077$ for S1 vs. $0.4753 - 0.4630 = 0.012$ for S2.

- Why best split is not necessarily the one with most balance child sizes : CART minimises the weighted impurity of the children, which depend on how pure the children are, not on how equal their sizes.

A3. Continuous attributes as a search problem.

- Chosen label : $N = 7$: 4 positive, 3 negative :

x	4	7	9	13	18	21	29
y	−	−	+	−	+	+	+

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Candidate cuts are midpoints where the label changes: between 7&9; 9&13; 13&18.

Cut	Left(P,N)	Right(P,N)	Weighted Gini
$x \leq 8$	(0,2), $G=0$	(4,1), $G=\frac{8}{25}$	$\frac{2}{7}\times 0 + \frac{5}{7}\times\frac{8}{25} = 0.2286$
$x \leq 11$	(1,2), $G=\frac{4}{9}$	(3,1), $G=3/8$	$\frac{3}{7}\times\frac{4}{9} + \frac{4}{7}\times\frac{3}{8} = 0.4048$
$x \leq 15.5$	(1,3), $G=\frac{3}{8}$	(3,0), $G=0$	$\frac{4}{7}\times\frac{3}{8} + \frac{3}{7}\times 0 = 0.2143$

The cuts clearly differ in quality: $x \leq 11$ (0.4048) much worse than $x \leq 8$ (0.2286) and $x \leq 15.5$ (0.2143)

- How a search tree compare them: sort the feature values. Generate candidate thresholds. For each candidate, compute the weighted impurity of the two children. Choose the threshold with lowest weighted impurity. The tree does the same for every feature at every node & keep the best pair.

A4. Pruning and missing values

- Reduce-error pruning: The subtree makes 7 errors on 30 validation examples; a single leaf makes 6. Replacing the subtree by a leaf does not increase validation error and gives a simpler tree. So prune: replace the subtree by a leaf. The extra branches were fitting noise in training data.
- Two strategies for a missing value at a split.

1. Surrogate splits (CART): use a correlate feature. Drawback: extra cost, fail if no correlated feature.
2. Fractional weights (C4.5): send down all branches by proportion. Drawback: several leaves => slower, less interpretable.

A5. Assignment bridge: tree audit

Setup: subject-disjoint split (21 train subject, 9 test subject), seed 42), scikit-learn decision tree classifier

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	Unconstrained	max_depth = 6
Maximum depth	18	6
Number of leaves	163	31
Training accuracy / macro-F1	1.0000 / 1.0000	0.9434 / 0.9412
Test accuracy / macro-F1	0.8622 / 0.8595	0.8544 / 0.8487

- Top 3 features (Gini importance): (1) tGravityAcc-min: 0.2399; (2) fBodyAccJerk-bandsEnergy()-1,16 (0.2013) (1)-X (3) angle(Y, gravityMean) (0.1316). Together about 57% of total impurity reduction.

- Prediction before running: reducing max depth to 6 will cut the leaves to at most 64; training accuracy will fall to 93-95% because more bias. Test accuracy will stay the same because less noise is fitted (85-87%).

- Comparison with actual result:
  + Leaf count dropped from 163 → 31 (~81%), within prediction
  + Training accuracy: 100% → 94.34%, inside predicted range.
  + Test accuray: 86.22% → 85.44% within prediction.

So the deep branches added very little real generalisation. There was no gain from pruning on the test set. Overall the depth-6 tree is much simpler and faster to inspect for nearly the same accuracy.

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